

Homework for Friday

Use Euclid's Lemma in each of the following proofs about irrational numbers.

1. Prove $\sqrt{p}$ is irrational if p is any prime.
2. Prove that $\sqrt{pq}$ is irrational if p and q are distinct primes.
3. Prove that $\sqrt[3]{10}$ is irrational.
4. Find the **multiplicative** inverse of each nonzero element in $\mathbf{Z}_5$ and $\mathbf{Z}_7$.

x	1	2	3	4
x^{-1}				

x	1	2	3	4	5	6
x^{-1}						

Pythagorean Theorem

5. Find **three** Pythagorean triples of the form $[60, b, c]$ with b and c greater than 60.
6. Use the index card I gave you in class to prove the Pythagorean Theorem.